

Wave 5X — Autonomous Commerce Intelligence (ACI)

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Mission

Transform GetPawsey into a self-learning ecommerce company that continuously discovers new revenue opportunities with minimal human intervention.

Architecture

Layer	Components
Edge functions	aci-orchestrator (8-stage)
Tables (18)	aci_settings, aci_runs, aci_run_steps, aci_audit_log, aci_budget_ledger, aci_approvals, aci_rollbacks, aci_r
Cron	aci-orchestrator-daily — 30 3 * * * UTC
UI	/admin/autonomous-commerce

Engines

- Market Intelligence (GSC, Pinterest trends, market trends)
- Competitor Intelligence (discovery, snapshots, gaps)
- Product Opportunity v2 (16 sub-scores, ranked)
- Revenue Intelligence (profit, margin, dead inventory)
- Trend Forecast (7/30/90/180/365 days, EWMA+linear)
- Recommendation Engine (ROI-prioritised)
- Task Generator (mode-aware, approval-gated)
- Learning Engine (approval feedback loop)

Live Run (simulation mode)

Step	Status	Metrics
market_intel	ok	{"signals": 50}
competitor_intel	ok	{"discovered": 0, "snapshots": 0}
opportunity_v2	ok	{"products": 428}
revenue_intel	ok	{"products": 428}
forecaster	ok	{"forecasts": 0}
recommender	ok	{"recommendations": 20}
task_generator	ok	{"tasks": 20, "approvals": 20, "mode": "simulation"}
learning	ok	{"approved": 0, "rejected": 0}

AI cost: \$0.00 • Cloud cost: \$0.00

Operator Guardrails

Setting	Value
mode_default	simulation
kill_switch	False
daily_ai_budget_usd	1.0
daily_cloud_budget_usd	2.0
max_tasks_per_day	50
autonomy_level	1

Production Readiness

Check	Pass
idempotent	✓
observable	✓
operator_controlled	✓
rollbackable	✓
rls_locked	✓
default_mode_safe	✓

Notes

- Mode initialised to 'simulation'. No production writes are performed without operator opt-in.
- Forecaster needs ≥ 5 days of `agp_signals_daily` to emit values.
- Competitor discovery seeded from `competitor_products`; live scraping deferred to follow-up wave.